1767–68

> **“** THE ELECTRIC SUBSTANCE WHICH SEPARATES THE **TWO CONDUCTORS,** POSSESSING THE TWO OPPOSITE KINDS OF ELECTRICITY, IS SAID TO BE **CHARGED. ”**

Joseph Priestley, English clergyman and scientist, from *The History and Present State of Electricity*, 1767

WHILE A CLERGYMAN IN LEEDS, FROM 1667–1773, JOSEPH PRIESTLEY experimented with chemistry and electricity, and published a highly successful book, ***The History and Present State of Electricity***, in 1767. Priestley noticed that **carbon dioxide**, or “fixed air” as it was then called, was a by-product of the fermenting beer vats of the local brewery. Experimenting with making fixed air himself, Priestley discovered that water impregnated with bubbles of carbon dioxide was pleasantly tangy. He passed on his discovery in the false hope that carbonated water could be used medicinally to prevent scurvy among sailors. In 1783, Swiss watchmaker Johann Jacob Schweppe (1740–1821) used the idea to launch the world’s carbonated drinks industry.

Italian naturalist **Lazzaro Spallanzani** (1729–99), the first to suggest that food might be preserved in airtight containers, discovered in 1767 that microbes are ever-present in the air and multiply naturally, instead of bursting into life spontaneously as was commonly believed. Most of the credit for this discovery would go to Louis Pasteur (see 1857–58), who demonstrated the same thing a century later.

On a more theoretical level, Leonhard Euler made the insightful suggestion that the **color of light is determined by its wavelength**.

Nooth’s apparatus
This apparatus was built by John Mervyn Nooth (1737–1828) in 1774 to make carbonated water for medicinal purposes in the way suggested by Priestley.

1769–70

French engineer Nicolas-Joseph Cugnot built his steam carriage to haul guns, but it was far too heavy to be practical, and may have crashed into a wall as this old print shows.

30,000
THE NUMBER OF PLANT SPECIMENS **JOSEPH BANKS** DISCOVERED IN **BOTANY BAY**

THE YEAR 1769 SAW THE CREATION OF MACHINES that would lead to the development of two key technologies. One could be called the forerunner of the automobile: the **steam-carriage** built by **French military engineer Nicolas-Joseph Cugnot** (1725–1804). The second, full-scale version of Cugnot’s carriage, built in 1770, had three wheels and a large copper boiler hanging over the front wheel. This machine could run for only 20 minutes and was so heavy and so impossible to steer or stop that it is said to have run out of control and demolished a wall.

The other revolutionary breakthrough was the **powered factory machine**, developed by **Richard Arkwright** (1732–92). It is often said that Arkwright’s **spinning machine**, which he commissioned clockmaker John Kay to build in 1769, was his greatest invention. This could automatically spin strong cotton yarn from frail English thread. Arguably Arkwright’s most inspired idea was to install scores of these machines in a specially built factory, where they were driven not by manpower but by water wheels—hence they were known as **water frames**. Arkwright’s revolutionary water-frame mill opened at Cromford on the Derwent River in Derbyshire, England, in 1771.

Across the other side of the world, British naval officer **Captain James Cook** (1728–79) was commanding HMS *Endeavour* on the first of his great voyages of discovery. He and his crew became the first Europeans to encounter the eastern coastline of Australia. On board was botanist **Joseph Banks**, who found 30,000 specimens of plant life, 1,600 of them unknown to European science, when they landed in a harbor later called Botany Bay. On June 3, 1769, Cook and his crew observed the **transit of Venus**—the passage of the planet in front of the Sun—from Tahiti. The event was also witnessed by astronomers all over the world. That year, French astronomer **Charles Messier** (1730–1817) made the first record of the vast cosmic cloud known as the **Orion Nebula**, a region in the constellation of Orion where new stars are formed.

Botany Bay
This 18th-century map of Botany Bay was engraved from charts made by James Cook during his exploration of Australia’s east coast.

Comparative sizes
Venus is tiny compared with the Sun, so when it is observed in transit the planet appears as just a small dark dot crossing the face of the Sun.

1767 English chemist Joseph Priestley invents **carbonated water**

1767 Italian naturalist **Lazzaro Spallanzani** challenges view that microbes generate spontaneously

1768 Leonhard Euler proposes that the color of light depends on wavelength

March 4, 1769 French astronomer Charles Messier first records the **Orion Nebula**

June 3, 1769 Transit of Venus observed worldwide, and seen by James Cook in Tahiti

1769 British inventor Richard Arkwright’s **water-powered spinning frame** built by John Kay

1769 French engineer **Nicolas-Joseph Cugnot** builds his steam carriage

1770 James Cook’s expedition lands in Botany Bay, Australia